

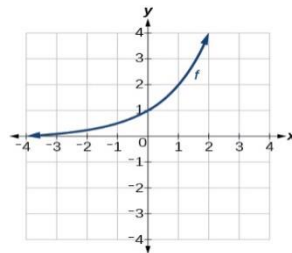
Section 1.5 Exercises

1. Write a formula for the function obtained when the graph of $f(x) = \sqrt{x}$ is shifted up 1 unit and to the left 2 units.
2. Write a formula for the function obtained when the graph of $f(x) = |x|$ is shifted down 3 units and to the right 1 unit.
3. Write a formula for the function obtained when the graph of $f(x) = \frac{1}{x}$ is shifted down 4 units and to the right 3 units.
4. Write a formula for the function obtained when the graph of $f(x) = \frac{1}{x^2}$ is shifted up 2 units and to the left 4 units.

Directions: For exercises 5 - 14, describe how the graph of the function is a transformation of the graph of the original function f .

5. $y = f(x - 49)$
6. $y = f(x + 43)$
7. $y = f(x + 3)$
8. $y = f(x - 4)$
9. $y = f(x) + 5$
10. $y = f(x) + 8$
11. $y = f(x) - 2$
12. $y = f(x) - 7$
13. $y = f(x - 2) + 3$
14. $y = f(x + 4) - 1$

Directions: For exercises 15 - 17, use the graph of $f(x) = 2^x$ shown below to sketch a graph of each transformation of $f(x)$.



15. $g(x) = 2^x + 1$
16. $h(x) = 2^x - 3$
17. $w(x) = 2^{x-1}$

Directions: For exercises 18 - 21, sketch a graph of each transformed function.

18. $f(t) = (t + 1)^2 - 3$

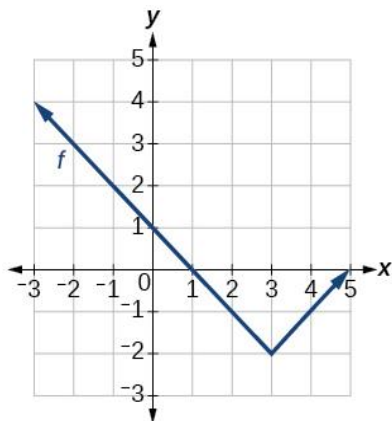
19. $h(x) = |x - 1| + 4$

20. $k(x) = (x - 2)^3 - 1$

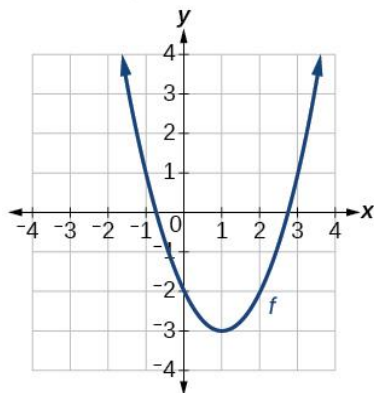
21. $m(t) = 3 + \sqrt{t + 2}$

Directions: For exercises 22 - 28, write an equation for each transformed function.

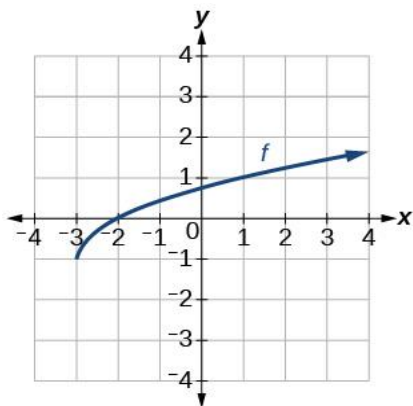
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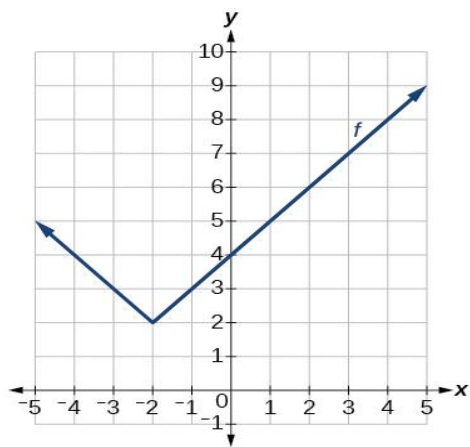
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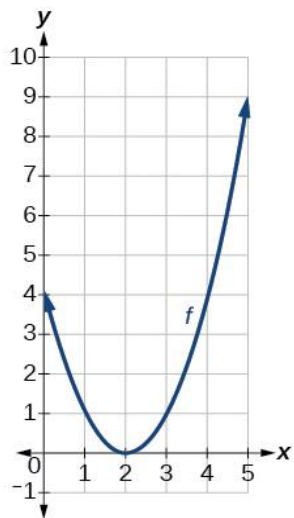
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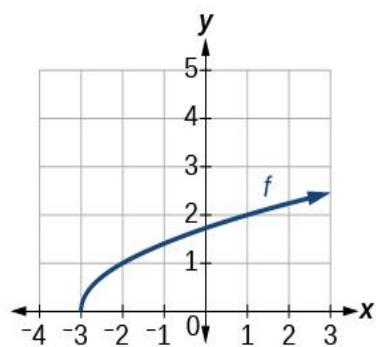
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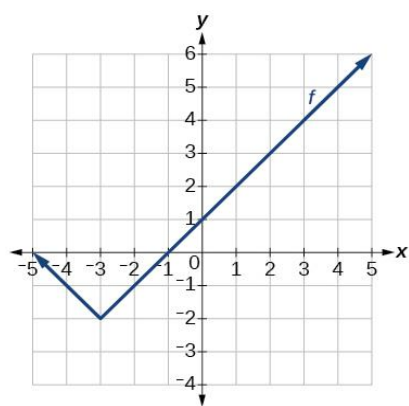
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27.



28.



Directions: For exercises 29 - 34, determine whether the function is odd, even, or neither.

29. $f(x) = 3x^4$

30. $g(x) = \sqrt{x}$

31. $h(x) = \frac{1}{x} + 3x$

32. $f(x) = (x - 2)^2$

33. $g(x) = 2x^4$

34. $h(x) = 2x - x^3$

Directions: For exercises 35 - 44, describe how the graph of each function is a transformation of the graph of the original function f .

35. $g(x) = -f(x)$

36. $g(x) = f(-x)$

37. $g(x) = 4f(x)$

38. $g(x) = 6f(x)$

39. $g(x) = f(5x)$

40. $g(x) = f(2x)$

41. $g(x) = f\left(\frac{1}{3}x\right)$

42. $g(x) = f\left(\frac{1}{5}x\right)$

43. $g(x) = 3f(-x)$

44. $g(x) = -f(3x)$